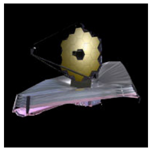

2391 - The resolved properties of PAHs at low metallicity

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
MIRI IC1613				
	2		MIRI Imaging	(1) IC-1613
	6		NIRCam Imaging	(1) IC-1613
SEXTANS-A				
	5		MIRI Imaging	(2) SEXTANS-A
	9		MIRI Imaging	(2) SEXTANS-A
	8		MIRI Imaging	(2) SEXTANS-A
	7		NIRCam Imaging	(2) SEXTANS-A

ABSTRACT

The NIR-MIR SED of galaxies is dominated by spectral features from polycyclic aromatic hydrocarbon (PAH) molecules fluorescing in regions illuminated by FUV photons. In low metallicity systems ($Z < 0.2$ Zo), previous studies with Spitzer have revealed a substantial deficiency in the

emission and abundance of PAHs, the origin of which remains unexplained due to the lack of deep, resolved, multi-band photometry or spectroscopy. We propose to obtain NIRCam and MIRI imaging to observe the 3.3, 7.7, and 11.3 micron PAH features in dusty star-forming regions in two low metallicity galaxies, IC 1613 (15% solar) and Sextans A (8% solar). The observations will allow us to 1) estimate the PAH properties, particularly their abundance, ionized fraction, and size distribution; and 2) correlate those properties with other tracers of the ISM, such as the star formation rate density, the distance and UV spectrum of massive stars, the distance to PAH-producing stellar sources, the ISM column density of gas and dust, and gas-phase abundances. By observing how the PAH properties and abundance vary with environment, we will constrain the mechanisms responsible for the formation, destruction, excitation, and low abundance of PAHs in those poorly shielded low metallicity environments. Since JWST will measure star formation rates as low as 10 Mo/yr from PAH emission at the peak of cosmic star formation at redshift z~2 when the average metallicity of the universe was about 10% solar, it is crucial that we deepen our physical understanding of PAHs at low metallicity. This proposal includes a joint HST 3 orbit request to map the dust extinction required to normalize the PAH abundance in IC 1613.

OBSERVING DESCRIPTION

We propose to obtain NIRCam and MIRI imaging in dusty star-forming regions in two low metallicity Local Group galaxies, IC 1613 (15% solar) and Sextans A (8% solar). The filters (F115W, F150W, F200W, F300M, F335M, F360M, F560W, F770W, F1000W, F1130W, F1500W) were selected to cover the PAH features at 3.3, 7.7, and 11.2 microns, as well as sample the SED in between, which includes a significant contribution from starlight that needs to be estimated and removed.

We will obtain single fields centered on the dustiest star-forming regions where PAH emission was previously detected with Spitzer. We will use the 4-point-sets dithers optimized for extended sources with MIRI, with starting-set=5 and number of sets = 1 (in the default positive direction) so that the pattern starts in the center of the array, which is in turned centered on the star-forming region. For NIRCam, we will use the INTRAMODULEX pattern with 4 dithers to maximize coverage and efficiency.

Exposure times were computed to reach a S/N of 1.5 per pixel for MIRI and 0.5 per pixel for NIRCam. We will bin the observations to 1" pixels to reach a S/N of 15 at 10 pc scales. This will allow us to used band ratios to derive the resolved properties of PAHs with good statistical significance.

For the MIRI observations, we will use 100 groups per integration (no saturation issues). For NIRCam, we will use 5-10 groups per integration (no saturation issues).

The observations for IC 1613 are split in two visits (1 MIRI, 1 NIRCam). The observations for Sextans A, which have longer exposure times due to the fainter PAH emission, are split in 4 visits (3 MIRI visits, 1 NIRCam visit) to allow schedulability.

All observations are background limited, as shown by the significant (>5%) difference in S/N between the low and high background ETC calculations. Therefore, we have added the corresponding special requirement in all observations.

Proposal 2391 - Targets - The resolved properties of PAHs at low metallicity

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	IC-1613	RA: 01 05 1.4616 (16.2560900d) Dec: +02 09 11.66 (2.15324d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES				
(2)	SEXTANS-A	RA: 10 11 6.4000 (152.7766667d) Dec: -04 42 17.66 (-4.70491d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Dwarf irregular galaxies] Extended=YES				

Observation

Proposal 2391, Observation 2

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	IC-1613	RA: 01 05 1.4616 (16.2560900d) Dec: +02 09 11.66 (2.15324d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F560W	FASTR1	100	5	1	Dither 1	4	20	5594.481	64089
2	F770W	FASTR1	100	3	1	Dither 1	4	12	3352.248	64089
3	F1000W	FASTR1	100	14	1	Dither 1	4	56	15684.526	64089
4	F1130W	FASTR1	100	10	1	Dither 1	4	40	11200.061	64089
5	F1500W	FASTR1	100	15	1	Dither 1	4	60	16805.642	64089

Special Requirements

Background Limited. Background no more than 20th percentile above minimum

Observation

Proposal 2391, Observation 6

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	IC-1613	RA: 01 05 1.4616 (16.2560900d) Dec: +02 09 11.66 (2.15324d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W	F300M	MEDIUM8	5	2	8	4	4165.867	64089
2	F115W	F360M	MEDIUM8	5	2	8	4	4165.867	64089
3	F200W	F335M	MEDIUM8	10	2	8	4	8460.575	64089

Special Requirements

Offset 82.0 arcsec, 0.0 arcsec
Background Limited. Background no more than 20th percentile above minimum

Observation

Proposal 2391, Observation 5

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SEXTANS-A	RA: 10 11 6.4000 (152.7766667d) Dec: -04 42 17.66 (-4.70491d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F560W	FASTR1	100	7	1	Dither 1	4	28	7836.713	64089
2	F770W	FASTR1	100	8	1	Dither 1	4	32	8957.829	64089
3	F1000W	FASTR1	100	25	1	Dither 1	4	100	28016.804	64089

Special Requirements

Background Limited. Background no more than 10th percentile above minimum

Observation

Proposal 2391, Observation 9

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SEXTANS-A	RA: 10 11 6.4000 (152.7766667d) Dec: -04 42 17.66 (-4.70491d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F1130W	FASTR1	100	30	1	Dither 1	4	120	33622.385	64089

Special Requirements

No Parallel Attachments
Background Limited. Background no more than 10th percentile above minimum

Observation

Proposal 2391, Observation 8

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SEXTANS-A	RA: 10 11 6.4000 (152.7766667d) Dec: -04 42 17.66 (-4.70491d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	4-Point-Sets				5	1	EXTENDED SOURCE	POSITIVE	DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F1500W	FASTR1	100	30	1	Dither 1	4	120	33622.385	64089

Special Requirements

Background Limited. Background no more than 20th percentile above minimum

Observation

Proposal 2391, Observation 7

Thu May 04 23:03:14 GMT 2023

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	SEXTANS-A	RA: 10 11 6.4000 (152.7766667d) Dec: -04 42 17.66 (-4.70491d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf irregular galaxies]
Extended=YES

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F150W	F300M	MEDIUM8	5	2	8	4	4165.867	64089
2	F115W	F360M	MEDIUM8	5	2	8	4	4165.867	64089
3	F200W	F335M	MEDIUM8	10	2	8	4	8460.575	64089

Special Requirements

Offset 82.0 arcsec, 0.0 arcsec
Background Limited. Background no more than 20th percentile above minimum